

AKAR KAUNG (He/Him/His)

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EDUCATION

M.Sc. in Computer Science, 2023

University of Minnesota, Twin Cities, MN

GPA: 3.78/4.00

Relevant Coursework: Intelligent Robotic Systems, Parallel Animation & Planning in Games, Computer Organization, Advanced Algorithms and Data Structures, Software Engineering II

B.Sc. in Computer Science, 2021

University of Minnesota, Twin Cities, MN

GPA: 3.59/4.00

Relevant Coursework: Machine Architecture and Organization, Introduction to Operating System, Development of Secure software system, Software Engineer I, Computer Network

RESEARCH EXPERIENCES

University of Minnesota - Twin Cities

May 2022 – Dec 2022

Research Assistant (Prof. Mattia Fazzini's Lab)

- Collaborated with PhD students to develop tools to streamline data preprocessing and analysis workflows for Natural Language Processing (NLP) models research, enhancing overall efficiency.
- Contributed to an automated project creation system to facilitate course-related assignments and improve reproducibility for Prof. Fazzini's course.
- Conducted bug pattern analysis to predict future software defects, applying data analysis techniques to enhance software quality assurance processes.

Independent Research Projects

• Analysis of Cloth Simulation in Parallel Programming

Jan 2023 – May 2023

Prof. Pen-Chung Yew

Researched and implemented parallel programming techniques to optimize computational efficiency of cloth simulations. Evaluated different approaches to improve performance and achieve smoother real-time rendering of complex physical interactions.

• Optimized and Efficient Path-Finding Algorithms

Sept 2022 – Dec 2022

Professor Nikolaos Papanikolopoulos

Conducted comparative analysis of pathfinding algorithms in simulated restaurant environments to identify optimal strategies for robotics navigation, aiding decision-making for operational improvements and customer experience enhancement.

WORK EXPERIENCES

Edirq Inc.

Jan 2025 – Present

Co-Founder & Software Engineer

- Reduced infrastructure downtime by 30% by designing and deploying a scalable AWS architecture (EC2, RDS/PostgreSQL, SES, SNS) with isolated dev/stage/prod environments, automated backups, and cross-region failover.
- Accelerated deployment by 80% via CI/CD pipelines (GitHub Actions) that included linting, unit testing, staging deploys, and rollback workflows.
- Improved system security and modularity by implementing role-based access, input validation, and fault-tolerant Node.js APIs backed by PostgreSQL.
- Maintained cross-platform API consistency by collaborating with React Native and React teams to define shared data contracts.
- Guided long-term technical direction through engineering roadmaps, design docs, and internal scalability reviews.
- Built a product-market fit (PMF) analytics system to track user signups, session data, feature usage, and DAU/MAU metrics, enabling data-driven growth decisions.

- Evaluated emerging technologies and selected optimal tools to meet business and technical goals.

Meta - Contractor

Mar 2025 – May 2025

Software QA Tester

- Validated performance, usability, and stability of Meta's AR/VR products by executing 500+ hands-on test cases across Oculus Quest 2, 3, 3S, Meta Ray-Bans, and unreleased prototypes—primarily targeting AI-driven features.
- Identified and documented 50+ high-impact bugs in software, AI behavior, and system performance through functional, exploratory, and usability testing—accelerating issue resolution and improving product quality.

Arch Gateway - Contractor

Dec 2024 – Mar 2025

Web Developer

- Oversaw all technical operations for the organization's technology team, including administration of G Suite and internal tools to support streamlined communication and collaboration.
- Designed, developed, and hosted the company website with a focus on education content, performance, usability, and responsive design.
- Implemented SEO strategies to boost visibility and drive organic traffic, resulting in improved search rankings and increased user engagement.

SuperTech FT

Feb 2024 – May 2024

Software Engineer Internship

- Enhanced the full-stack educational software suite, facilitating real-time communication between a React frontend and a Flask backend deployed on Raspberry Pi.
- Redesigned and maintained the organization's website using JavaScript, HTML, and CSS to ensure modern functionality and aesthetics.

Outlier AI

Jan 2024 – Dec 2024

AI Prompt Engineer

- Designed and engineered prompt solutions to improve system performance and reliability, yielding an 80% approval rate for outputs flagged as high-quality responses.
- Analyzed prompt behavior and failure cases by reviewing peer submissions, identifying systemic weaknesses, and applying targeted refinements to improve reliability and robustness.

Green River College

May 2018 – Jun 2019

International Student Ambassador

- Collaborated with cross-functional teams, including student organizations, administrative staff, and faculty members, to ensure comprehensive event planning and seamless coordination of logistics and resources.
- Acted as a liaison between international students and college administration, advocating for student needs and promoting inclusive campus initiatives to enhance the overall student experience.
- Provided guidance and support to international students, facilitating their integration into campus life and promoting cross-cultural understanding and friendship among the student body.

TEACHING EXPERIENCES

University of Minnesota - Twin Cities

Jan 2022 – May 2023

Graduate Instructor

- Taught core software engineering topics—including Agile methodologies, Git version control, unit testing, and design patterns in C++—to junior and senior undergraduate students.
- Led individual and group troubleshooting sessions to address diverse software issues and bugs, fostering student confidence and technical proficiency.
- Created instructional materials and lab content for over 1,000 students, emphasizing practical application of theoretical concepts.
- Supervised 30+ teaching assistants over multiple semesters, coordinating grading, managing course logistics, and responding to student inquiries.

University of Minnesota - Twin Cities

Sept 2021 – Jan 2022

Teaching Assistant (Prof. Mattia Fazzini)

- Worked closely with Professor Mattia Fazzini to analyze and refine lab and assignment instructions, contributing to the enhancement of the learning experience

- Led lab discussion sessions to a class of 105 undergraduate students, facilitating their progress and ensuring successful completion of assignments and projects.
- Conducted thorough evaluations of student performance through project assessments, offering constructive feedback to promote continuous improvement and a deeper understanding of course concepts.

LEADERSHIP EXPERIENCES

Green River College - Auburn, WA

CORE Leader/Organizer, Campus Life

May 2018 – Jun 2019

- Organized and led welcome overnight camping trips for over 300 new international students every quarter.
- Managed logistics, including transportation, food, volunteers, budgeting, and student coordination.
- Interviewed, trained, and supervised volunteers to support event operations and enhance student engagement.

Vice President, Green River Burmese Club

Dec 2017 – Jun 2019

- Organized cultural events and community-building activities to support Burmese students at Green River College.
- Collaborated with faculty and student organizations to promote cultural awareness and inclusivity.
- Provided leadership and guidance to club members, fostering a supportive and engaged student community.

Event Organizer, Asian Students Union (ASU)

Dec 2018 – Jun 2019

- Planned and coordinated cultural events to celebrate Asian heritage and foster community engagement.
- Managed event logistics, including venue booking, budgeting, and marketing, to ensure successful execution.
- Collaborated with student organizations and faculty to promote diversity and cultural awareness on campus.

Decoration Department Lead, Lunar New Year Party

Feb 2019

- Led the decoration team in designing and setting up festive Lunar New Year-themed decorations for the event.
- Managed volunteers, budget, and materials to create an immersive and culturally authentic atmosphere.
- Collaborated with the college art department to design and craft DIY decorations.

Security Department Lead, Halloween Party

Dec 2018

- Led the volunteer security team to ensure student safety and smooth event operations during the Halloween Party.
- Collaborated with college security officers to handle potential threats and enforce safety protocols.
- Coordinated crowd management and assisted attendees while addressing minor security concerns.

Web Master, International Student Events (ISE)

Jun 2017 – May 2019

- Designed, developed, and maintained the official website for International Student Events (ISE) to enhance event visibility and engagement.
- Managed website content, ensuring up-to-date information on events, resources, and student activities.

Gusto College, Yangon, Myanmar

2016

President - Student Government Board

- Led the Student Government Board in representing student interests and organizing campus initiatives.
- Coordinated events, meetings, and projects to enhance student engagement and academic support.
- Acted as a liaison between students and faculty, advocating for student needs and campus improvements.

VOLUNTEER AND COMMUNITY ENGAGEMENT

Panelist Speaker, Education USA

Jul 2019

Yangon, Myanmar

- Provided prospective college students with insights on cultural adaptation, time management, and academic success while studying abroad.
- Advised students on what to expect in a new educational system and strategies for a smooth transition to international studies.

Volunteer, City Forest Clean Up

May 2018

Auburn, WA

- Supported environmental preservation efforts by cleaning up litter and maintaining the health of local forests and green spaces.

Volunteer, Phi Theta Kappa Honor Society

Mar 2017 – Dec 2017

Green River College, Auburn, WA

- Took part in a community service initiative to remove graffiti and revitalize several neighborhoods.

Volunteer, GRC Welcome Week
Green River College, Auburn, WA

Feb 2017

- Helped new students transition to college life by leading campus tours, organizing activities, and facilitating engagement through interactive games.

Volunteer, GRC CORE Trip
Green River College, Auburn, WA

Aug 2017

- Led student groups during an overnight camping trip, facilitated team-building activities and helped new students adjust through storytelling and mentorship.

Volunteer, Water Aid Donation
Tayat, Myanmar

Mar 2014

- Assisted in distributing clean water supplies to communities affected by drought and helped raise awareness about water scarcity and conservation efforts.

HONORS, AWARDS, AND ACHIEVEMENTS

1. **Dean's List of Scholars, College of Science and Engineering, University of Minnesota, January 2020 – May 2021.** Awarded to students whose grade point average is 3.66 or above and who completed 12 hours of graded coursework per semester.
2. **Campus Life Leadership Award, Campus Life, Green River College, June 2019.** Recognized for outstanding leadership and contributions as an International Student Ambassador, fostering community engagement and supporting fellow students.
3. **Honorable Mention for Best Decoration Event, The Current, Green River College, June 2019.** Recognized for creating a vibrant and culturally authentic atmosphere, featuring traditional lanterns, calligraphy, and symbolic decorations that captured the essence of the Lunar New Year celebration.
4. **Student Representative of Math Division Tenure Committee, Green River College, January 2019 - June 2019.** Served as one of the few selected student representatives on the Math Division Tenure Committee, contributing to faculty tenure evaluations and decision-making.
5. **Student Representative of IT Budget Committee, Green River College, May 2018 - June 2019.** Served as the sole student representative on the IT Budget Committee, providing input on technology funding decisions and advocating for student needs in IT resource allocation.
6. **Awarded to participate in Council of Unions and Student Programs - Student Leadership Conference, Green River College, May 2018.** Awarded the opportunity as a selected student from Green River College to participate in the Council of Unions and Student Programs - Student Leadership Conference, representing the college in leadership development and advocacy discussions.
7. **Certificate of Recognition for Thesis Defense, American Mathematical Association of Two-Year Colleges, Green River College, June 2018.** Recognized for successfully presenting a thesis defense in the 2018 AMATYC Student Research League, an exclusive competition for selected participants chosen by the Green River College Mathematics Division.
8. **Recognition of Outstanding Achievement in Mathematics, Green River College, June 2018.** Awarded MAA membership by the Green River Mathematics Division for exceptional academic performance and excellence in mathematics.
9. **Best Design Award ChatBot Hackathon, Myanmar, November 2016.** Awarded \$750 for developing a chatbot with optimized UI/UX and multilingual support, enhancing user engagement and efficiency.

Projects

Space-Warfare ([URL](#))

Tech Stack: UE5, C++, Blender

- Created smart spaceship simulation in UE5, predicting and engaging enemy spaceships while implementing evasion tactics.
- Designed a self-destruct feature for the spaceship, triggered when health drops below a threshold.
- Integrated advanced evasion tactics into the spaceship's AI algorithms, enabling it to successfully dodge and outmaneuver enemy spaceships when detected and targeted.

Back2Town ([URL](#))

Tech Stack: Unity, C#, Blender

- Developed a 3D character simulation within Unity, incorporating advanced NPC movement using the Dijkstra pathfinding algorithm.
- Designed and implemented a sophisticated collision avoidance system to enhance gameplay realism and character interactions within the virtual environment.
- Utilized Unity's tools to bring detailed, realistic environments to life, improving both visual appeal and functionality.

Autonomous Drone Transportation Simulation ([URL](#))

Tech Stack: C++, HTML, CSS, Javascript, CSS, UML, Agile

- Developed a simulation for autonomous passenger drones, including algorithmic control of drone flight and passenger transport functionality.
- Produced detailed UML diagrams and comprehensive documentation, supporting code optimization and system architecture clarity.
- Refined existing codebase using Factory, Decorator, and Strategy design patterns to improve code organization and structure.

Cloth And Fluid Simulation ([URL](#))

Tech Stack: Processing

- Developed a cloth simulation and fluid dynamics model using processing to simulate realistic cloth movement and interactions with fluid, achieving smoother simulations.
- Integrated collision detection to ensure realistic interactions between the cloth and surrounding objects, improving the overall realism of the simulation.
- Designed and integrated a computational model for real-time visualization, demonstrating realistic cloth behavior and fluid interactions in dynamic environments.

Core NLP XML To Brat Ann Conversion ([URL](#))

Tech Stack: Python, XML, ANN

- Developed a Python script to convert Core NLP XML files to Brat Ann format, handling Part-of-Speech (POS) tagging and dependency relations based on Enhanced++ Dependencies.
- Implemented functions for parsing XML structures, extracting token information, and generating annotation and configuration files, ensuring compatibility with Brat visualization tools.
- Utilized XML parsing techniques in Python's ElementTree library to automate the conversion process, improving efficiency and reducing manual annotation work.